

Personal Information



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- Bachelor / 2012.9-2015.6 / Jilin Engineering Normal University / Computer Science and Technology (CHSI Verifiable)
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Highlights

- **Career Positioning:** Technical Director / Full Stack Architect. Successfully delivered multiple products and architectures from 0 to 1 and 1 to 100.
- **AI & Algorithms:** Proficient in enterprise-grade RAG intelligent dialogue systems, search & recommendation engines, and related algorithms. In-depth practice in large-scale vector retrieval (ES + Milvus), recall, and ranking strategies.
- **Search / Vector Retrieval:** Design of ElasticSearch and Milvus retrieval and indexing, supporting large-scale recall and ranking.
- **Tech Stack:** Primarily Java/Golang/Python/PHP, complemented by Vue/TypeScript. Focus on Backend, AI Algorithms, Distributed Systems, and Microservices.
- **Architecture Experience:** Microservices / Distributed Systems / High Availability Multi-Active, K8S Cluster Management, OpenResty Gateway Rate Limiting & Disaster Recovery, Hybrid Cloud (AliCloud + IDC).
- **Performance & Stability:** MySQL Indexing & R/W Splitting, Sharding, Redis Caching, Message Compensation, Sync to Async conversion, and Parallelization.
- **High Concurrency:** Front-end peak clipping based on OpenResty Gateway & Redis; Backend support via Async Decoupling (RabbitMQ/Kafka), R/W Splitting, and Connection Pool extension.
- **HA & Multi-Active:** Canary / Rolling Release strategies to ensure stability.
- **DevOps & O&M:** Docker, CI/CD, CMDB, K8S, Zabbix/Grafana Monitoring, ES Log Platform, GoAccess Log Analysis.
- **Security & Reliability:** Self-developed WAF/Security Strategies, DDOS/CC Protection experience, Data Consistency & Idempotency Design.
- **Team Management:** Built and managed teams of 10-50 people from scratch. Experience in Recruitment, Performance Management, Process Standardization, and Engineering Efficiency Improvement.
- **Business & Delivery:** Iteration of toC platforms, implementation of toB/toG projects (Research Plagiarism Check Intranet Delivery, Cross-border E-commerce).

- **Collaboration & Drive:** Systems thinking and result-oriented. Smooth cross-departmental coordination, capable of working independently, high stress tolerance, and self-driven learning.

Work Experience

Hangzhou Weilan Sports Technology Co., Ltd. | Technical Director | 2023.11 - 2025.09

Peak Team Size: 20, Average: ~10 **Key Projects:**

1. Moment Sports (WeChat Mini Program / H5 / APP, 0 to 1)

- Built and led a 15-person Product/R&D/Design team, streamlining Requirements, Design, R&D, Testing, and Launch processes.
- Deconstructed business domains, designed overall technical architecture & governance (Spring Boot, Nacos, ES, Redis), and implemented DevOps, Canary Release, and Monitoring/Alerting.
- Independently completed core backend and partial frontend development, built High Availability & Disaster Recovery systems, ensuring stable operation post-launch.
- Formulated technical budget and personnel planning, driving company qualifications, compliance, and process implementation.

2. Peking Union Medical College - Institute of Medical Information · Paper and Image Similarity Check

- Managed business docking and solution review, organizing data and permission boundaries according to research scenarios.
- Architecture & Deployment: Intranet Delivery, Data Security Isolation, CI/CD, and O&M Monitoring.
- Text/Image Vectorization, Similarity Comparison for Plagiarism Check, supporting Large-scale Library Retrieval.
- Organized delivery demonstrations and training, completing intranet implementation and acceptance.

3. Mydesign Cross-border E-commerce Independent Site

- Communicated requirements with stakeholders, co-produced product prototypes, and iterated incrementally by module.
- Built cross-border data synchronization links and task orchestration, handling differences in exchange rates, time zones, taxes, and languages.
- Automated O&M (AWS + DevOps), optimized domestic access to overseas links and acceleration.
- Improved First Screen Loading and Key Transaction Metrics, stably handling overseas traffic.

Shanghai Lianyin Network Technology Group Co., Ltd. | Full Stack Architect | 2022.11 - 2023.11

Peak Team Size: 24, Average: ~14 **Key Projects:**

1. Pangxie Game Account Trading Network

- Independently completed the architecture and service deployment of Pangxie 1.0, transforming it from a standalone unit to a Multi-Active High Availability Distributed Architecture.

- Perfectly supported 8x business growth.
- Optimized frontend and backend technical architecture, reducing technical costs and improving Technical ROI.
- Planned and designed multi-person, multi-environment collaborative development, improving development efficiency and system robustness.

2. Java-based Pangxie 1.0 Mall Big Data Search

- Built and developed the search system from 0 to 1 based on ES. Achieved incremental synchronization by reading binlog asynchronously to ES on top of full MySQL data synchronization.
- Implemented Game Account Recall, Filtering, Fine Ranking, and Business Re-ranking based on User Personas and User Behavior Data, improving search accuracy and relevance.

3. Organized & Managed O&M and Big Data Team

- Managed technical resources for 104 developers across Pangxie Game Account Trading Network, Pangxie Game Mall, Pangxie Esports Platform, Innovation Division, and IM Customer Service System.
- Completely withstood a single peak of 600GB DDOS and 15G Bandwidth CC Hybrid Attacks.

4. Assisted Technical Director in Cost Control & Organizational Management

- Assisted in organizational management and project flow management of the technical team, growing from 18 to 104 people.
- Assisted in personnel flow, code quality, and document management across project groups.

Hangzhou Taoyou Technology Co., Ltd. | Technical Director | 2020.6 - 2022.6

Peak Team Size: 48, Average: ~30 **Key Projects:**

1. E-commerce Platform Development from 0-1 based on Golang

- Used Gin framework, gRPC communication, built Microservices Architecture. Modules split into User, Trading, Payment Management (including 3rd party payment), Log modules, etc.
- Used OpenResty, Redis, Golang to build High Concurrency scenarios, successfully completing multiple Flash Sale events.

2. Java Refactoring of Juxiangyou PHP Project

- Refactoring and optimization of a platform with 500k DAU and 3000 QPS. Refactored User, Finance, and Background Script services using Java, improving project robustness and maintainability.
- Used Docker-Compose for service orchestration, unified development environments, doubling the speed of horizontal machine scaling.

3. Juxiangyou Redeployment & Architecture Adjustment, Building HA Multi-Active Architecture

- Hidden IP, built DDOS Defense Layer.
- OpenResty + Golang as Gateway, implementing Downgrade, Rate Limiting, Business Layer Protection, and CC Attack Defense.
- Withstood hybrid attacks of up to 400G DDOS and 5G Bandwidth CC.
- Upgraded Captcha to Puzzle Captcha.
- Deployed SQL Audit, Bastion Host, Zabbix Log Monitoring, DingTalk Robot, and SMS Alarms, increasing architecture hierarchy.

Hangzhou Tianzuan Network Technology | Technical Lead | 2018.11 - 2020.6

Peak Team Size: 8, Average: ~5 **Key Projects:**

1. Tiaotiaozhu PC and H5 Project

- Developed from 0 to 1. After project formation, built a 5-person team, mainly responsible for Core Code Development and Server Stability.
- Docker-Compose service orchestration, achieving Low-cost Rapid Expansion.
- Built High-Performance Architecture. Intermediate forwarding layer used PHP + OpenResty, Core Business Logic developed using Golang. Mixed use of HTTP and gRPC protocols ensured development speed while improving performance.

Other Work:

1. Handled various difficult problems and technical direction decisions.
2. Integrated SDKs from third-party companies like Alibaba and Tencent.
3. O&M Automated Deployment.
4. Managed work plans, performance, bonuses, code control, team building, and recruitment for a 5-person technical team. Improved motivation and controlled R&D costs.

Horgos Tuomei Entertainment | PHP R&D Engineer | 2015.8 - 2018.11

Key Projects:

1. Online Earning Projects

- Maintenance of Web Game Trial Module.
- Lead Programmer for Chess Module, completed Chess Module and all related activities from 0 to 1.
- Script maintenance for Platform Entertainment Games.

2. Gem Planet Project (APP)

- Mining Service, generating revenue daily according to algorithms.
- Backend for Cocos games like Pipile, Farm, etc.

Other Work:

1. Integrated SDKs from third-party companies like Alibaba and Tencent.
2. Optimized and adjusted MySQL.

Project Experience

2023.11 - 2025.09 | Hangzhou Weilan Sports Technology Co., Ltd. | Technical Director

Project Description: Moment Sports

Role: Technical Director

Technology: SpringBoot 2.7, Python, Golang, MyBatis-Plus, MyBatis, MySQL, Redis, Nacos, RabbitMQ, Consul, Elasticsearch, Vue3, TS, Vite, UniApp, Docker, OpenResty, Ansible, etc.

Content: Completed Moment Sports WeChat Mini Program, H5, and APP from 0 to 1.

- Designed infrastructure based on business models and technical budgets, tracked project progress, and coordinated technical resources across departments.
- Advanced from 0 to 1 and 1 to 100, built a complete technical team, responsible for Technical Recruitment, Performance Management, and Technical Budget Formulation & Management.
- Dynamically determined requirements, distributed and tracked tasks, strictly controlling project rhythm and quality.
- Formulated and implemented R&D standards and processes (Frontend/Backend Code Standards, Middleware, CDN, OSS usage standards).
- Built and transformed O&M infrastructure: Evolved from Alibaba Cloud to Alibaba Cloud + IDC Hybrid Cloud.
- Implemented Canary/Rolling Release and Distributed Traffic Governance via OpenResty, reducing production risks.
- Built Automated CI/CD, CMDB, and Internal Wiki; GoAccess for Nginx Log Analysis; Zabbix/Grafana Monitoring; Elasticsearch Log Retrieval.
- Completed core backend development for Users, Accounts, Event Registration, Team Tournament Registration, Merchant Printer Services, etc.
- Completed development of H5 pages and UniApp Mini Programs.
- Optimized for High Concurrency: MySQL Indexing & R/W Splitting, Sharding, Redis Cache Design, Sync to Async & Parallel Reads, Connection Pool Optimization.
- Introduced MQ Compensation Mechanism to ensure data consistency and timely delivery of order messages to club merchants.

Project Description: Peking Union Medical College - Institute of Medical Information · Paper/Image Plagiarism Check

Role: Technical Director

Technology: SpringBoot 2.7, Python, MyBatis-Plus, MySQL, Redis, RabbitMQ, Vue3, TS, Vite, Docker, OpenResty, Milvus, ResNet50, CLIP, DenseNet121, Swin Transformer, DINOv2, etc.

Content: Paper/Image Similarity Check for Research Scenarios

- Docked with toG requirements, output technical solutions and budget quotations, produced prototypes using Axure.
- Deconstructed milestones and scheduling, handled task distribution, tracking, and progress control.
- To reduce computing costs, penetrated to local server room via frp and local three-line BGP high-bandwidth lines.
- Built Automated CI/CD via OpenResty Canary/Rolling Release.

- Deployed and managed MySQL, Redis, RabbitMQ, Milvus; designed and maintained indexes.
- Implemented Global PDF Parsing, engineered algorithm implementation, achieved index design and retrieval for 5 million small image vectors.
- Organized delivery demonstrations and training, completed intranet environment deployment and acceptance.

Project Description: Mydesign Cross-border E-commerce Independent Site

Role: Technical Director

Technology: SpringBoot 2.7, Python, MyBatis-Plus, MyBatis, MySQL, Redis, Nacos, RabbitMQ, ElasticSearch, Vue3, TS, Vite, UniApp, Docker, OpenResty, Ansible, etc.

Content: Independent Site from Prototype to Overseas Growth

- Docked with stakeholders, co-produced product prototypes and roadmaps.
- Split business modules, planned project cycles, adopted Agile Rolling Iteration.
- Built Technical Architecture and AWS Deployment, O&M Automation, and Cross-border Data Synchronization.
- Optimized Domestic Access to Overseas Network Links and Acceleration.
- Implemented Engineering of Personalized Search & Recommendation, and participated in partial algorithm implementation.

2022.11 - 2023.11 | Shanghai Lianyin Network Technology Group Co., Ltd. | Full Stack Architect

Project Description: Pangxie Game Account Trading Network

Role: Full Stack Architect

Technology: Docker, OpenResty, ThinkPHP, SpringBoot, ElasticSearch, Zabbix, etc.

Content: Independently completed Pangxie 1.0 architecture and service deployment. Assisted Technical Director in improving service quality, code quality management, and personnel organization management.

- Re-architected the service from a standalone unit (Business side 1 server, Admin side 1 server, QPS 30, DAU 5000) to a Multi-module, Multi-tier, Multi-environment High Availability Architecture.
- Sorted out and extended PHP business logic, re-architected large resource transmission and Websocket services, reducing technical costs and improving service stability.
- Sorted out Frontend Vue2 framework, organized component libraries, adjusted security strategies, and optimized business-side key acquisition security.
- Organized refactoring of Customer Service IM Client, Server, IM Message Service, and Customer Service System. Completed two versions, iteratively improving load capacity.
- Adjusted and planned business-side project team members, managed and controlled Code Quality, CI/CD Standards, Git Standards, O&M and Development Standards, etc.
- Re-sorted and simplified the Search System. Used SpringBoot to complete Game Account Recall, Filtering, Fine Ranking, and Business Re-ranking, reducing RT by 80%.
- Implemented Scheduled Tasks using Golang, ensuring task stability and reliability.
- Implemented Canary/Rolling Release and Distributed Traffic Governance using OpenResty, reducing production risks.
- Used Canal to implement MySQL binlog synchronization to Elasticsearch, achieving large text search functionality via Elasticsearch.
- Used RabbitMQ to implement Async Decoupling and Async Message Processing, while splitting Cold/Hot Data Sources to improve data reading performance.

2021.3 - 2021.6 | Hangzhou Taoyou Technology Co., Ltd. | Technical Director

Project Description: Quandoumai E-commerce

Role: Technical Director

Technology: Gin, gRPC, Protobuf, Nacos, Consul, Elasticsearch (Microservices)

Content: Developed Mall from 0-1, completed all work from deployment architecture to final delivery.

- Design and Code Standards for Gin framework middleware, toolsets, and project framework.
- Module splitting: User Module, Trading Module, Payment Management Module (including 3rd party payment), Log Module, etc.
- Used Nacos, Consul, Kafka, gRPC, etc., for Microservice Governance, Discovery, and Communication.
- Used Docker-Compose to orchestrate services, achieving service consistency and rapid horizontal expansion.
- Automated Compilation and Deployment based on GitLab-Runner O&M.
- Troubleshooting and resolution of difficult problems like Golang Memory Overflow and Full-link Troubleshooting.
- Multiple Flash Sale activities, using OpenResty, Redis, Golang to build High Concurrency scenarios.

2020.6 - 2021.12 | Hangzhou Taoyou Technology Co., Ltd. | Technical Director ****Project Description****:

Juxiangyou Refactoring

****Role****: Technical Director

****Technology****: PHP, Java, SpringBoot 2.7, MyBatis-Plus, Redis, Docker-Compose, etc.

****Content****: Refactored Core Modules of PHP Project using Java. - SpringBoot refactoring of PHP project core modules: User, User Finance, Background Service Scripts. - Implemented gRPC and HTTP interaction between PHP project and Java project. - Transformed KVM Virtual Machines to Docker-Compose orchestration. - Automated Compilation and Deployment using GitLab-Runner. - Handled difficult problems like PHP Process Fake Death.

2020.6 - 2022.6 | Hangzhou Taoyou Technology Co., Ltd. | Technical Director

Project Description: Juxiangyou Architecture Optimization & Adjustment

Role: Technical Director

Technology: OpenResty, Lua, Golang, Java, Redis, Docker, etc.

Content: Built High Availability Multi-Active Architecture, improving project robustness.

- Hidden IP, built DDOS Defense Layer.
- OpenResty + Golang as Gateway, defending against CC Attacks.
- Implemented Downgrade, Rate Limiting, achieving protection for the Business Layer.
- Withstood hybrid attacks of up to 400G DDOS and 5G Bandwidth CC.
- Upgraded Captcha to Puzzle Captcha.
- Deployed SQL Audit, Bastion Host, Zabbix Log Monitoring, DingTalk Robot, and SMS Alarms.

2018.11 - 2020.6 | Hangzhou Tianzuan Network Technology | Technical Director

Project Description: Tiaotiaozhu Project

Role: Technical Director

Technology: OpenResty, ThinkPHP, Docker-Compose, Golang, Redis, etc.

Content: Tiaotiaozhu PC and H5 Project

- Developed Tiaotiaozhu PC project from 0 to 1 using ThinkPHP.
- Built a 5-person team, developed Tiaotiaozhu H5 project and Golang refactoring.
- Docker-Compose service orchestration, ensuring service consistency and reducing maintenance difficulty.
- Used OpenResty as Gateway, PHP for Business Development, Script Services developed using Golang.
- Used Redis to complete multiple Red Packet Rain activities.
- Completed Automated O&M using GitLab-Runner.

2017.1 - 2018.11 | Horgos Tuomei Entertainment | PHP Development Engineer

Project Description: Gem Planet Project (APP)

Role: PHP Development Engineer

Technology: Yii2, MySQL, Redis, RabbitMQ, etc.

Content: Gem Planet Project (APP)

- Used Redis to complete Mining Service for millions of cold users, generating revenue daily according to algorithms.
- Used RabbitMQ to asynchronously calculate Mining Service for 700k DAU hot users, calculating revenue.
- Backend for Cocos games like Pipile, Farm, etc.

2015.8 - 2017.1 | Horgos Tuomei Entertainment | PHP Development Engineer

Project Description: Online Earning Projects

Role: PHP Development Engineer

Technology: ThinkPHP, MySQL, Redis, etc.

Content: Online Earning Projects

- Maintenance of Web Game Trial Module.
- Completed Chess Module and all related activities from 0 to 1.
- Backend componentization of Chess Management, making Chess Ad updates simple and easy to understand.
- Script maintenance for Platform Entertainment Games.

Solved Technical Challenges

- **Withstood 400G DDOS and 5G Bandwidth CC Hybrid Attacks**
 - Adjusted architecture, differentiated based on user models.
 - Monitored server performance metrics, set alarms, and early warnings.
 - Strengthened log review, customized WAF.
- **Mongo Replica Set Sharding Master-Slave Arbitration Docker Containerization & Full Data Sync**
 - Docker-Compose orchestration of Sharding Replica Sets, forming One Master, Multiple Slaves, One Arbiter.
 - Full migration of 10T data.
- **Distributed High Availability Load, Nginx Process Cannot Restart/Kill**
- **Imitation Geetest Puzzle Captcha**
 - [PHP Version](#), Production used [Golang Version](#) Puzzle Captcha.
- **PHP Process Massive Fake Death Troubleshooting**
 - PHP-FPM parameter tuning, process count adjustment.
 - MySQL Slow Query troubleshooting.
 - Xhprof analysis.
- **Seamless Migration of Massive MySQL Data**
 - Full Migration: Batch dump of data.
 - Default used MySQL tools.
 - Used go-dumper when dump exceptions occurred.
 - Incremental Migration:
 - Read binlog, import data into Message Queue (Kafka), asynchronously import data.
- **Accelerated Domestic Access to Overseas Servers**
- **Business Service Migration & Expansion**
 - Docker image orchestration and container management for rapid deployment.
 - GitLab-Runner for Automated O&M.
- **HTTPS Inbound/Outbound SSL Version Issues**

Open Source Projects

- [AI Paper Plagiarism Check](#) Demo: <https://seven.weilantech.com/>
 - Provides batch upload detection for PDF documents and images. Supports intra-document, strategy, and full-library check modes. Integrates SIFT, SURF, and other feature point matching algorithms for image similarity analysis. Implements a three-level expert review process and automatically generates standardized detection reports.
- [Captcha-Pic](#)
 - Puzzle Captcha made with Golang, HTML version and API interface version.
- [Laradock Secondary Development](#)
 - [laradock](#) is an open source project for PHPers. On this basis, through modification and improvement of Docker image management, it is made more suitable for use in production environments. Added OpenResty, OpenResty-Waf, OpenResty-Lor, VeryNginx, Golang, Python, Walle, Xhgui, Xhprof, GitLab-Runner, Filebeat.
- [webcron](#) (<https://github.com/Allen-LPL/webCron>)

- Developed with Golang language and Beego framework. Features permission management, email notification, fuzzy task query, and task termination control.

Technical Articles

- [Long Connection Management of IM Project](#)
- [Deep Dive into Golang Map Expansion Source Code](#)
- [Golang Slice Expansion Mechanism & Source Analysis](#)
- [Distributed Storage HDFS Architecture Principles & Practice Notes](#)
- [ETL Tool Kettle Data Processing Practice Notes](#)

Tech Stack

- **Primary Languages:** Java, Python, PHP, Golang, Lua, Vue/TypeScript
- **Frameworks:** Spring Boot, ThinkPHP, Yii, Hyperf, Beego, Gin
- **RPC:** gRPC, Protobuf
- **Data Storage:** MySQL, Redis, MongoDB, Memcache, go-cache, BigCache, Milvus
- **Retrieval/Search:** Elasticsearch (incl. Filebeat+Grafana link)
- **Messaging & Async:** RabbitMQ, Kafka
- **Gateway & WAF:** Nginx, OpenResty (incl. self-developed WAF strategies)
- **Container & Orchestration:** Docker, Docker Compose, Kubernetes
- **CI/Configuration:** GitLab Runner, Webhook, Ansible, CMDB
- **Monitoring & Observability:** Zabbix, Grafana, Elasticsearch Log Platform, PProf
- **HA & Registry:** Nacos, Consul, HA, Keepalived
- **Project Governance:** Project Plan/Docs/Code Standards/Defect Tracking/Automated O&M/Production Environment Isolation & Audit
- **Algorithms:** SIFT, SURF, CLIP, DINOv2, Swin Transformer, ResNet50, DenseNet121